Candidate Prompt: FDE Build, Demo & Delivery!

Before You Begin

Congratulations on making it to the **FDE Build, Demo & Delivery** panel interview. We're continuing the movie theater scenario — the customer is ready for a redesign and wants the FDE in the room to show what a real solution looks like. Great FDEs read between the lines and design production-ready solutions that scale — not just answer the question asked. You will showcase this in this interview.

- **The Prompt & Time-Boxed Build:** You have 4 hours to create an updated architecture that addresses the current gaps and build a POC that demonstrates the critical new components.
- **Real-World Simulation:** After the build period, you will present in a 60-minute presentation. You'll role play an FDE briefing a panel of senior stakeholders on a proposed architectural extension to an existing solution.

  The Persona Panel: Interviewers will role play specific stakeholder roles (Engineering / Platform Lead, CIO / CDO, VP of Theater Operations). Lead the room, manage the clock, and handle pushback on feasibility, cost, timelines, and architectural trade-offs.

Scenario Prompt

You are an FDE invited to present a technical solution to the customer's issues with the current operational observability and proposed recommendation engine. Today, the theater managers can review their theater's operation KPIs in a dashboard.

Customer Pain Points: In previous discussions with the customer, the team has noted several pain points in the current architecture:

- **Data quality is silently absorbed by the people closest to the data.** Managers have surfaced quality issues in the joined datasets, but the dashboard gives them no way to correct or annotate what they see. They've adapted by

mentally working around the bad rows — which means the "ground truth" only lives in their heads, every decision carries a mental tax, and trust in the platform is eroding.

- **Governance isn't built into the view.** Every user sees the same dashboard regardless of role. The business wants access scoped by level of leadership — so theater managers, regional leaders, and executives each see only what they're supposed to — while still giving leaders a way to roll up across theaters and surface outliers from a more global view.
- **The dashboard reports the past; it doesn't shape the future.** It shows sales KPIs but offers no recommendations on *where* and *when* to place movies within a theater for best sales outcomes. Managers are left to make scheduling and placement decisions on instinct, and the platform isn't helping them turn data into a better business outcome.
- **The serving layer is read-only when the business needs to act.** Managers and executives want to do more than view metrics — they need to log and annotate data quality issues, accept or override recommendations, and trigger downstream actions from the same place they see the data. A static dashboard pushes that work out into spreadsheets, email, and side channels; the business wants an interactive surface where decisions and follow-ups happen in-product.

Your Assignment:

1. **Proposed Architecture: Propose an end-to-end architecture that addresses all of the issues above**. You are expected to design and present the full end-to-end proposal and discuss the business value add of everything in the design, not only what POC you will demo.

2. **Build a POC. Pick <u>one</u>:**
    - **Tightened data engineering**: Build data quality checks into the pipeline that detect and surface issues (rather than silently passing them through), and design a pattern for corrections or annotations to flow back into the governed dataset. Demonstrate how the pipeline handles bad data and how a user's feedback loop works.
    - **A recommendation engine**: an AI/ML model tracked in MLFlow that recommends schedules in movie-screen-time format.
    - **A deployed Databricks App**: an interactive, role-aware serving layer that replaces the current dashboard.

An in-depth working version of one POC option will count more than light touch on all three - prioritize a strong delivery on what is feasible in the 4 hour build window.

You will need to make some assumptions for your design. Make sure to call out these during the delivery and be ready to speak to your thought process behind the assumptions.

Your Starting Point

Your starting assets: FDE Zip Folder. The current state architecture and artifacts are deployable into your Databricks Free workspace via DAB. Please review the README.md in the zip for a full description of what is included as well as instructions on deployment to the workspace and setup.

Deliverables for Your Session

The Delivery portion of the interview is weighted 50%. Make sure to lead with the business outcome of your proposal and use your future state architecture and POC as supporting evidence. You will be asked questions from both technical and non-technical personas - be ready to discuss your solution to all audiences.

Bring the following into your 60-minute customer-facing session:

- **Future-state architecture diagram:** the extension grounded in the existing architecture; what stays, what changes, what's deferred.
- **Live POC demo:** The critical path runs. Walk through the running artifact(s), narrate what the user (e.g., theater manager) can now do, and explain the advantage over the current architecture.
- **Prioritized action plan:** top 3–5 ranked actions with rough impact vs. effort, quick wins vs. deeper work, and the decisions the customer needs to make in this meeting vs. defer.
- **Any additional artifacts:** anything else needed for presenting (notebooks, slides, scratch diagrams, etc.)

Who's in the Room

This is a technical architecture review, not a sales meeting. Read the room and tailor your narrative to the audience:

Persona	What They're Focused On
VP of Theater Operations *Wants to know when the business sees value*	Does this actually help a theater manager do their job? Will managers trust and act on recommendations? Clear line from capability to profitability? Adoption plan?
CIO / CDO *Speaks in strategy, cost, and timelines*	Strategic fit with the broader data/AI direction? Cost to build, run, and scale? Realistic timeline to value? Governance for the new capability? Sustainable without FDE hand-holding?
Engineering / Platform Lead *Wants technical soundness and a clear blast radius*	Are the architectural choices sound? Does the extension fit the existing architecture or create tech debt? What's the blast radius on current pipelines? Production-ready or prototype that needs rewriting?

Tips for Build Success

- Ground the extension in the existing architecture – don't redesign from scratch
- Use AI as a force multiplier but make sure you understand the solution
- Prioritize for impact and feasibility in your build

Suggested 4 Hour Build Block

Time	Focus
15 minutes	Prompt and scenario review: constraints, scoring emphasis, anchors (business outcome, architecture extension, POC); decide what "done" looks like for your prep
15 minutes	Existing architecture review + ranked gaps; draft proposed extension
2 hours	Build: POC critical path, future-state diagram, supporting artifacts
30 minutes	Action plan reasoning and optional asset creation
30 minutes	Rehearse opening, narrative, objections
30 minutes	Dry-run demo end-to-end

Tips for Delivery Success

- **Lead with the business outcome,** not the pipeline or catalog structure.
- **Be specific** about the architecture choices and how they solve the business problems.
- **Show, don't tell** — let the architecture and POC anchor your delivery

Suggested Interview Time Blocks

Time	Focus
5 minutes	Open the meeting, set the agenda, frame the problem and impact to the business.
30 minutes	Walk through current-state review, proposed extension, POC demo, action plan.
15 minutes	Panel Q&A and objection handling
10 minutes	Debrief / Wrap-Up: drop the personas — discuss what you'd do differently, how you used AI during the build

Use AI Tools Freely

There are no restrictions on AI tooling usage. The scaffold FDE Zip Folder includes README.md, SETUP.md, and FAQ.md which can be used to help your AI tool gather context.

"AI as a force multiplier" is an **explicit signal** in this interview. Use LLMs, code assistants, and prompt engineering to accelerate architecture design, POC build, and supporting artifacts (diagrams, narrative, demo data). In the debrief, be ready to articulate where AI helped, where you intervened, and approximately how much you used.

Using Databricks Free Edition & Hit a Limit? If you hit a limit during your build, you can claim a Credit Boost to keep working, **click here** for more details. It's there if you need it, but you don't need to use it to do well. Plenty of strong builds never touch it. Focus on your approach, not on maxing out compute.

Evaluation Rubric

The rubric for this interview is weighted **50% for the Architecture & POC and 50% on the Project Delivery**. Key elements to both are below:

- **Business value:** Outcomes that matter to the panelists — not technical capability for its own sake.

- **Technical judgment:** Sound choices grounded in the existing architecture; appropriate Databricks primitives; trade-offs explained.
- **AI-builder mindset:** AI used visibly as a force multiplier; candidate can explain where AI helped and where they intervened.
- **POC quality:** A running prototype that makes the extension concrete.
- **Leading the room:** Agenda, clarity under pressure, driving decision without panic.
- **Stakeholder management:** Engineering feasibility, CIO cost / timeline pressure, business adoption — calibrate depth to the persona asking.
- **Objection handling:** Evidence over reassurance; backbone with humility.

Finally, if you have any issues during the build window or questions, please reach out to your recruiter.
We're excited to see how you approach this problem!